SAI SANTOSH CHEETHIRALA

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CS sophomore at PES University who builds ML from the metal up — a neural-net trainer written in raw **WebGPU/WGSL compute shaders** (~97% MNIST, fully gradient-checked), a **GPT-2 mechanistic-interpretability** microscope, and a shipped local-first study PWA. Incoming research intern in RAG & LLM reliability at PESU Intelligence Labs.

PROJECTS

GlassBox — Mechanistic-interpretability microscope for GPT-2 small github.com/santoshcheethiralame-dot/GlassBox

Python · FastAPI · TransformerLens · PyTorch · scikit-learn · React / TypeScript

- Built an interactive dashboard over a FastAPI / TransformerLens service that visualizes attention, activation patching, per-layer linear probes, residual-stream PCA, and neuron analysis for GPT-2 small (124M)
- Localized factual recall to exact (layer, position) sites via causal tracing — activation patching swings the answer's logit difference from **+1.89 to -2.81**
- Stopped probe leakage with StratifiedGroupKFold on source word; serialized concurrent forward passes behind a process-wide lock to keep activation-hook caches consistent

Gradient — Neural-net training engine that runs entirely on the GPU, in-browser santoshcheethiralame-dot.github.io/GRADIENT

TypeScript · WebGPU · WGSL · React 19 · Vite · Web Workers

- Wrote the **entire training stack as hand-authored WGSL compute shaders** — forward pass, backprop, Adam/SGD, and 16×16 tiled matmul — with no Python, server, or WASM
- Trained a 2-layer MLP to **~97% MNIST** test accuracy in a Web Worker, plus a from-scratch nano-GPT char transformer with hand-derived, gradient-checked backprop through softmax-attention and layer-norm
- Verified every kernel against an f64 CPU oracle with numerical gradient checks re-run on each load (**44/44 passing**, rel. err < 1e-3); added a CPU fallback reusing the same oracle

Orbit — Shipped local-first study planner (PWA) with applied ML orbit-virid-gamma.vercel.app

React 19 · TypeScript · Dexie.js / IndexedDB · OpenRouter API · Tailwind CSS · Vite

- **Built and shipped** an offline-first PWA (live on Vercel) with zero server dependency, **sub-5ms** local scheduling, and a reactive UI via Dexie useLiveQuery
- Designed the planning engine: exam-readiness scoring (Ebbinghaus volume curve × exponential forgetting decay), SM-2 spaced repetition, a learned productivity curve, and deadline back-scheduling with infeasibility warnings
- Wired OpenRouter free-model routing into an AI coach, exam generator, and cheat-sheet / flashcard tools; full JSON + iCal export

RESEARCH EXPERIENCE

PESU Intelligence Labs — Research Intern, RAG & LLM Reliability Jun – Jul 2026 (Incoming)

- Selected to research hallucination attribution in retrieval-augmented generation — a diagnostic layer that pinpoints which retrieved chunks introduce errors, replacing binary pass/fail with explainable, source-level failure analysis for high-stakes deployments

TECHNICAL SKILLS

Languages: Python, C, TypeScript, JavaScript, SQL, WGSL, HTML/CSS

ML & Research: PyTorch, TransformerLens, scikit-learn, NumPy, Pandas; mechanistic interpretability (activation patching, linear probes), spaced repetition

Web & Systems: React 19, FastAPI, Pydantic, Dexie.js / IndexedDB, WebGPU, Vite, Tailwind CSS, PWA, Docker, REST APIs

Tools: Git, Linux, Vercel, OpenRouter API

EDUCATION

PES University — B.Tech, Computer Science Engineering Sep 2024 – May 2028

CGPA 7.93 / 10 (top 25% cohort) · Academic Distinction Award (SGPA ≥ 7.75) — Sem 2 & Sem 4

Relevant Coursework: Data Structures & Algorithms, Design & Analysis of Algorithms, Operating Systems, Computer Networks, Linear Algebra, Mathematics for ML, Data Science

LEADERSHIP & ACTIVITIES

Sahay AI — Student AI Club, PES University · Multimedia Domain Head Oct 2025 – Mar 2026

- Led multimedia for a 50+ member club; built a content pipeline and posting schedule that drove **5–6× growth** in content views over 3 months

National-Level Air Rifle Competitor — 10m Air Rifle, National School Games · a precision event built on sustained focus and methodical calibration under pressure